

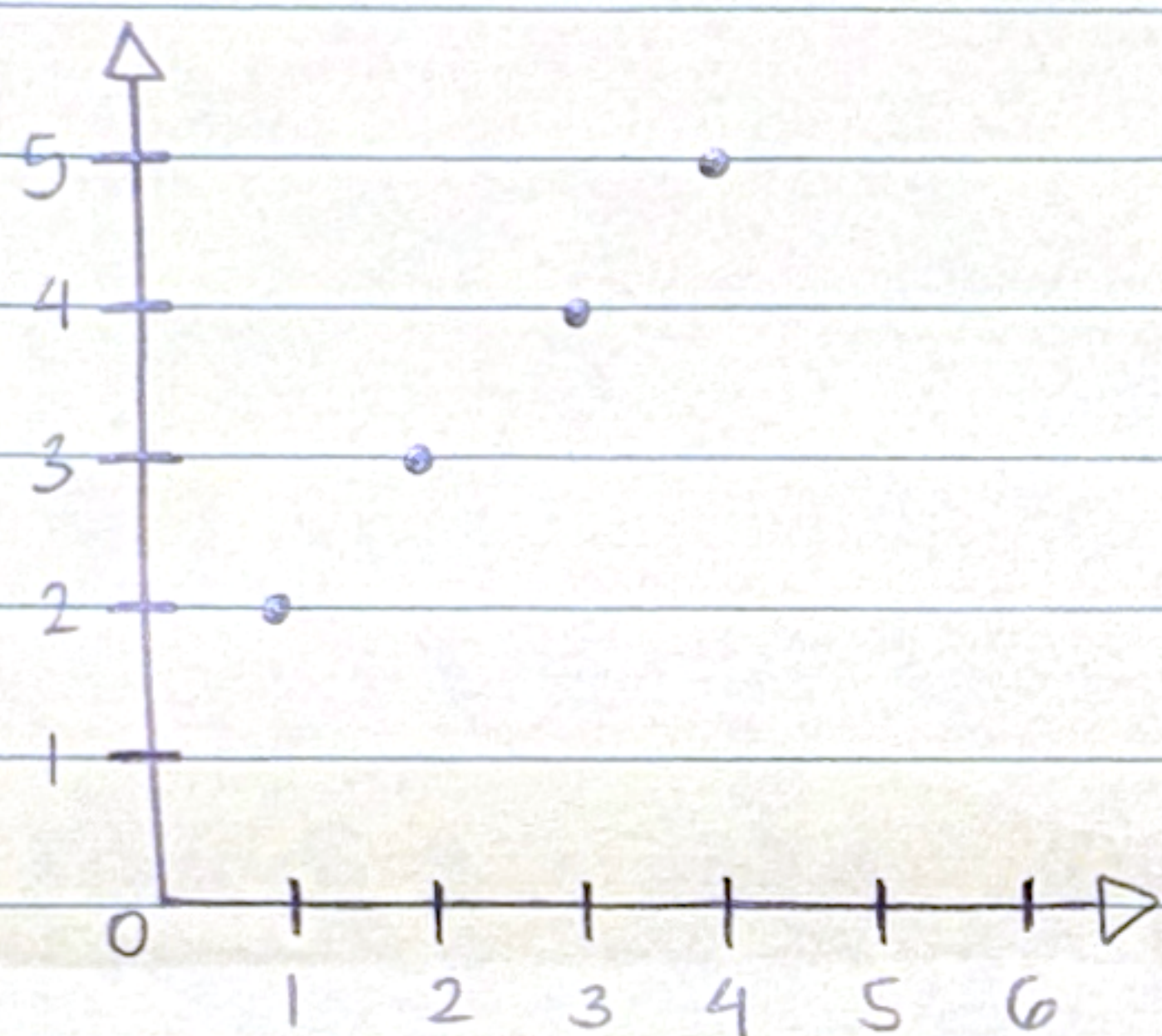
LEON MARCO A. DEVELA
COM232

ADVANCED MACHINE LEARNING
EXERCISE 4

Casino Customer Dataset

CUSTOMER	AVG BET PER VISIT	VISITS PER MONTH
A	1	2
B	2	3
C	3	4
D	4	5

① Plot the dataset on a 2-dimensional plane



② Compute the mean of each feature

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 3 \\ 3 & 4 \\ 4 & 5 \end{bmatrix}$$

$$\mu_1 = \frac{1+2+3+4}{4} = 2.5$$

$$\mu_2 = \frac{2+3+4+5}{4} = 3.5$$

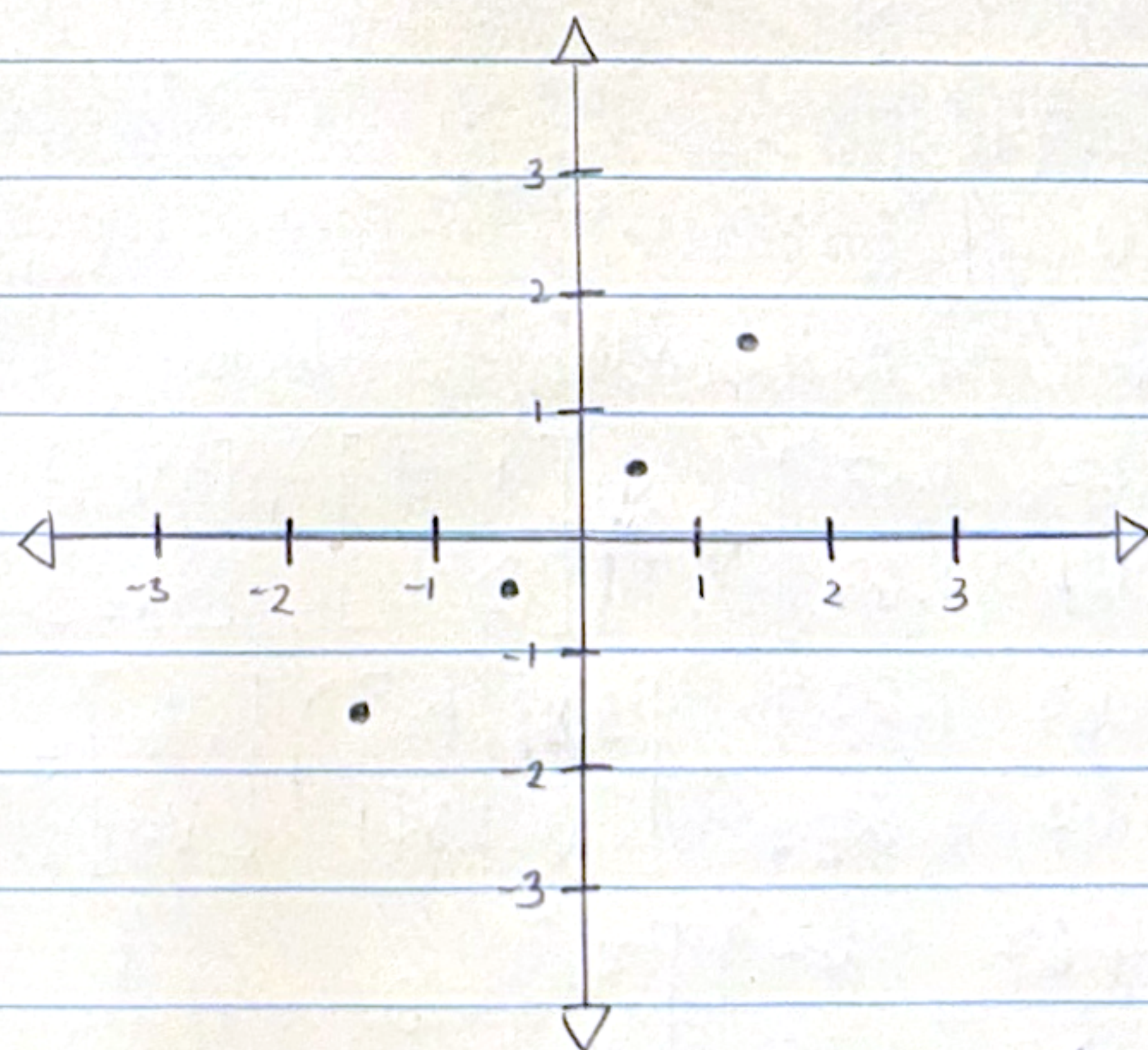
$$\mu = [2.5 \quad 3.5]$$

③ Center the dataset by subtracting the mean of each sample

$$A_c = \begin{bmatrix} 1-2.5 & 2-3.5 \\ 2-2.5 & 3-3.5 \\ 3-2.5 & 4-3.5 \\ 4-2.5 & 5-3.5 \end{bmatrix}$$

④ Write down the centered data matrix

$$A_c = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$



⑤ Compute the sample covariance matrix of the centered data

$$\Sigma = \frac{1}{4-1} \begin{bmatrix} -1.5 & -0.5 & 0.5 & 1.5 \\ -1.5 & -0.5 & 0.5 & 1.5 \end{bmatrix} \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix}$$

$$\Sigma = \frac{1}{4-1} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$$

$$\Sigma = \frac{1}{3} \begin{bmatrix} 5 & 5 \\ 5 & 5 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix}$$

⑥ Compute the eigenvalues of the covariance matrix

$$\det \left(\begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} - \lambda \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right)$$

$$\det \left(\begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} - \begin{bmatrix} \lambda & 0 \\ 0 & \lambda \end{bmatrix} \right)$$

$$\det \begin{vmatrix} 1.67 - \lambda & 1.67 \\ 1.67 & 1.67 - \lambda \end{vmatrix} = 0$$

$$= (1.67 - \lambda)^2 - 2.79 = 0$$

$$= (1.67 - \lambda)(1.67 - \lambda) - 2.79 = 0$$

$$= \cancel{2.79} - 1.67\lambda - 1.67\lambda + \lambda^2 - \cancel{2.79} = 0$$

$$= \lambda^2 - 3.34\lambda = 0$$

$$= (\lambda - 0)(\lambda - 3.34) = 0$$

$$\lambda_1 = 3.34$$

$$\lambda_2 = 0$$

⑦ Compute the corresponding eigenvectors

$$\left(\begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} - 3.34 \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\left(\begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} - \begin{bmatrix} 3.34 & 0 \\ 0 & 3.34 \end{bmatrix} \right) \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$\begin{bmatrix} -1.67 & 1.67 \\ 1.67 & -1.67 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = 0$$

$$= \begin{bmatrix} -1.67v_1 + 1.67v_2 = 0 \\ 1.67v_1 - 1.67v_2 = 0 \end{bmatrix}$$

$$= \begin{bmatrix} 1.67v_2 = 1.67v_1 \\ 1.67v_1 = 1.67v_2 \end{bmatrix}$$

$$= \begin{bmatrix} v_2 = v_1 \\ v_1 = v_2 \end{bmatrix}$$

$$V = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

⑧ Normalize the eigenvectors

$$U = \frac{V}{\|V\|} = \frac{1}{\|V\|} = \frac{1}{\sqrt{1^2 + 1^2}} = \frac{1}{\sqrt{2}}$$

$$V_{\text{normal}} = \frac{V}{\|V\|} = \frac{\begin{bmatrix} 1 \\ 1 \end{bmatrix}}{\sqrt{2}} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{bmatrix} = \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

⑨ Check if the eigenvectors and eigenvalues are correct using $AV = \lambda V$

$$\begin{bmatrix} 1.67 & 1.67 \\ 1.67 & 1.67 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = 3.34 \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

$$\begin{bmatrix} 3.34 \\ 3.34 \end{bmatrix} = \begin{bmatrix} 3.34 \\ 3.34 \end{bmatrix}$$

⑩ Identify the first principal component

ANSWER: $PC_1 = 3.34$

EXPLANATION: Since $\lambda_1 = 3.34 > \lambda_2 = 0$, the eigenvector associated with λ_1 represents the direction where the data is most spread out. It also captures the maximum variance in the data.

⑪ Project the centered data onto the first principal component

$$\text{new projection} = \begin{bmatrix} -1.5 & -1.5 \\ -0.5 & -0.5 \\ 0.5 & 0.5 \\ 1.5 & 1.5 \end{bmatrix} \begin{bmatrix} 0.707 \\ 0.707 \end{bmatrix}$$

$$\text{new projection} = \begin{bmatrix} -1.061 + -1.061 \\ -0.354 + -0.354 \\ 0.354 + 0.354 \\ 1.061 + 1.061 \end{bmatrix}$$

$$\text{new projection} = \begin{bmatrix} -2.122 \\ -0.707 \\ 0.707 \\ 2.122 \end{bmatrix}$$

12. Write down the resulting 1-dimensional representation of each sample

$$\text{new projection} = \begin{bmatrix} -2.122 \\ -0.707 \\ 0.707 \\ 2.122 \end{bmatrix}$$

